

SYLLABUS

1B01ICSC: INTRODUCTION TO C PROGRAMMING

COURSE OUTCOME

CO1: Aware about basics of programming.

CO2: Capable to analyze the problem and design algorithm and flowchart.

CO3: Familiar the basics of high-level language – C.

CO4: Able to develop efficient and error free programs in C.

Unit I:

Computer Programming and Languages: Introduction, Developing a Program, Program Development Cycle, Algorithm, Flowchart: Flowchart Symbols, Guidelines for Preparing Flowcharts, Benefits of Flowcharts, Limitations of Flowcharts, Examples of Algorithm and Flowchart. [Text Book 1]

(5 Hrs)

Unit II:

Overview of C: History of C, Importance of C, Basic Structure of C Programming Style, Executing a C program, Source Code, Object Code, Executable File, File Extensions, Character Set, C Tokens - Keywords, Identifiers, Constants, Variables, Data Types, Declaration of Variables, Assigning Values to Variables, Reading Data from Keyboard, Operators and Expressions: Arithmetic Operator, Relational Operators, Logical Operators, Assignment Operators, Increment and Decrement Operators, Conditional Operator, Bitwise Operator, Special Operators,. Arithmetic Expressions, Precedence of Arithmetic Operators, Type Conversion in Expressions.

(5 Hrs)

Unit III:

Managing Input Output Operation: Reading a Character, Writing a Character, Formatted Input, Formatted Output. Decision Making and Branching: Decision Making with if Statement - Simple if, if - else, Nested if - else, else if Ladder, switch Statement, go to Statement, Decision Making and Looping: while, do-while, for Statement, Jumps in Loops - break and continue Statements.

(4 Hrs)

Unit IV:

Arrays: Introduction, One Dimensional Arrays - Declaration of Arrays, Initialization of Arrays; Two-Dimensional Arrays - Initializing Two-Dimensional Arrays, Multi-Dimensional Array, Handling of Character Strings: Introduction, Declaring and Initializing String

Variables, Reading a Line of Text, Writing Strings to Screen, Arithmetic Operations on Characters, String Handling functions: strlen, strcpy, strcmp, strcat, strrev.

(4 Hrs)

Books for Study:

1. Introduction to information technology ITL Education solutions Limited, second Edition
2. Programming in ANSI C Second Edition – E Balagurusamy – Tata McGraw-Hill
Publishing company Limited

Books for Reference:

1. Let us C, Yeshavant Kanetkar, 16th Edn, BPB
2. Programming in C, Ashok N Kamthane, Pearson Education
3. Computer Basics and c Programming, V. Rajaraman, PHI, 2008 6
4. Fundamentals of information technology, Dr. S.B Kishor, A.S Khandelwal, 2nd Ed,
Published by DAS GANU Prakashan.

Online References:

1. <http://www.yspuniversity.ac.in/cic/algorithm-manual.pdf>
2. https://www.it.iitb.ac.in/~vijaya/ssrvvm/dokuwiki/media/s6_l7_20jan.pdf

Marks Including Choice:

Unit	Marks
I	14
II	14
III	16
IV	16